

# Example Worksheet: The Kitchen Sink

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**Video:** <https://www.youtube.com/watch?v=dQw4w9WgXcQ>

## 1 Prerequisites

Prerequisites, further reading, and anything else you want to say are just ordinary  $\text{\LaTeX}$  — no special markup. For this sheet:

- comfort with probability on finite sets ( $\mathbb{E}$ , distributions);
- one variable of calculus (we differentiate  $-t \log t$  once).

### What you'll learn

#### Using this template

- Recognise every markup element the  $\text{tex} \rightarrow \text{MDX}$  pipeline supports.
- Copy this file as the starting point for a new worksheet.

#### Structuring learning outcomes

- Group outcomes with  $\text{\subsection*{\{...\}}}$  plus an `itemize` when a flat list gets long.
- Drop the  $\text{\subsection*}$  headings for a short sheet – a single `itemize` is fine.

This dummy worksheet demonstrates every construct that survives the round trip from  $\text{\LaTeX}$  to the curriculum website. Compile it with `pdflatex` (twice, then `bibtex`, then twice again for the bibliography), or just read the source side by side with the rendered page.

## 2 Prose, math, and cross-references

Ordinary prose converts verbatim, including *emphasis*, **bold**, `monospace`, and special characters like 50% or §3. Inline math is copied byte-for-byte:  $e^{i\pi} + 1 = 0$ , and KaTeX handles comparisons like  $x_{<t}$  without trouble. Custom notation defined in the preamble just works:  $D_{\text{KL}}(p \parallel q) \geq 0$ ,  $\arg \max_a \mathbb{E}[R \mid a]$ , for  $p, q$  distributions on  $\mathbb{R}^n$ .

Display math works as

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

and as numbered equations you can reference:

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1}$$

Multi-line derivations use `align`:

$$(a+b)^2 = (a+b)(a+b) \tag{2}$$

$$= a^2 + 2ab + b^2. \tag{3}$$

Cross-references resolve to the exact text LaTeX prints: Equation (1) sits in Section 2; below we will meet Definition 3.1 and Theorem 3.2. Lists are fine too:

- first bullet;
- second bullet, with math  $2^n$ .

## 3 Definitions, theorems, and callouts

Label anything you want to reference. Any placement `\LATEX` binds correctly is valid (top level of the environment); right after the `begin` is clearest.

**Definition 3.1** (entropy). The *entropy* of a distribution  $p$  on a finite set  $\mathcal{X}$  is

$$H(p) = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

**Theorem 3.2** (Gibbs' inequality). For any distributions  $p, q$  on  $\mathcal{X}$ ,  $-\sum_x p(x) \log q(x) \geq H(p)$ , with equality iff  $p = q$ .

*Proof.* By definition,  $D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$  by Jensen's inequality applied to the concave function  $\log$ ; rearranging gives the claim. Equality in Jensen requires  $p(x)/q(x)$  constant, i.e.  $p = q$ . (On the website this proof is collapsible, like a solution.)  $\square$

**Lemma 3.3.**  $H(p) \leq \log |\mathcal{X}|$ , with equality iff  $p$  is uniform.

*Proof.* Apply Theorem 3.2 with  $q$  uniform. □

### Warning

Callouts flag pitfalls, tips, or side notes. Types: **note**, **tip**, **warning**. They may be labelled and referenced, like this one.

### A titled callout

A second optional argument titles the box: it replaces the “Tip” label on the PDF frame and heads the box on the web. Handy for a highlighted key equation,  $H(p) = -\sum_x p(x) \log p(x)$ , or a named aside.

### Remark

A **remark** is an aside; on the website it renders as a note-style callout. Referencing works: see Callout 3.1 above (and this remark is Remark 3.1).

## 4 Exercises and solutions

The optional argument of an exercise is its title, exactly as in `amsthm`. An optional `\important` mark (a ★ flagging the sheet’s most important exercises) goes right after the label. Subparts are a plain `enumerate`; label an item to reference it. Every solution *names its exercise* — `[ex:label]` is mandatory — so you may place solutions right after each exercise or collect them at the end, as you prefer. Exercises don’t require solutions; label any exercise a solution or reference points at.

**Exercise 4.1 (Entropy warm-up).** Let  $p$  be a distribution on a finite set  $\mathcal{X}$ .

- (a) Show that  $H(p) \geq 0$ .
- (b) For which  $p$  is  $H(p) = 0$ ?

**Exercise 4.2 (Gibbs, the hard direction).** (★) Prove the equality case of Theorem 3.2 directly, without citing Jensen’s equality condition. [Note: this one is solved at the end of the sheet instead of inline, to demonstrate both solution placements.]

Hints also have a block form: unnumbered, never labeled, rendered right where it is written — a bold lead-in on paper, a collapsible drop-down on the web.

**Hint:** Consider  $f(t) = t - 1 - \log t$ : it is nonnegative on  $t > 0$  and vanishes only at  $t = 1$ .

## 5 Figures and citations

Figures use `figure + \includegraphics` (export TikZ to PDF the same way; keep assets in `fig/`).

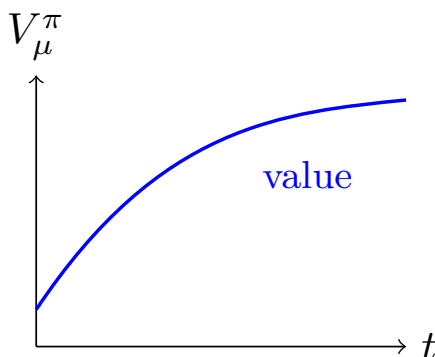


Figure 1: An example figure. On the web this becomes a **Figure** component.

See Figure 1. Citations come from the `.bib` file: the idea of entropy is due to [2]; for a textbook treatment see [1, ch. 4]. External links are plain [hyperlinks](#), and footnotes are supported.<sup>1</sup>

### Selected solutions

### References

- [1] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [2] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.

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<sup>1</sup>On the web this becomes a numbered footnote: the marker links down to a note at the foot of the page, and the note links back. In the PDF it is an ordinary L<sup>A</sup>T<sub>E</sub>X footnote.